GlucoTwin India: System Architecture

Hybrid physiological + ML digital twin for Type 2 Diabetes · static EHR stream × dynamic wearable stream → 2-hour adverse-event prediction + clinician what-if simulation

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Happiest Health Digital Twin Challenge 2026

1 · DATA SOURCES

STATIC STREAM · EHR

Demographics, diagnoses (SNOMED CT)
HbA1c history, FPG, lipids, eGFR, BP (LOINC)
Medications: metformin, SU, insulin (RxNorm)
Genetics: TCF7L2 rs7903146, polygenic risk

HL7 FHIR R4 bundles · ABHA identifier
Custom synthetic generator, FHIR R4 export;
CGMacros bloodwork for real-data validation

DYNAMIC STREAM · WEARABLES / IoT

Continuous glucose monitor (5-min)
Smartwatch: heart rate, HRV (RMSSD), steps
Sleep stages (light / deep / REM)
Meal log: 27-item Indian food library
(carbs, GI, fat, protein, fibre)

Apple Health / Google Fit-style JSON
Physiology-based synthetic generator + open
CGMacros (PhysioNet): CGM + Fitbit + meals
Replayed as a live stream (simulated clock)

VIRTUAL PATIENT ENGINE

Extended Bergman oral minimal model: dawn
phenomenon, sleep debt, stress, SU/insulin;
CGM sensor model (lag, AR(1) noise, dropouts)
80 Indian T2D / prediabetes patients × 14 days

2 · INGESTION & FUSION

FHIR parser → EHR feature vector

15 static features (age, BMI, HbA1c, FPG,
duration, therapy, TCF7L2, PRS, eGFR ...)

→ EHR-informed twin prior
Gb ← fasting glucose; insulin action ← HbA1c,
disease status

Stream alignment (5-min grid)

Causal gap fill (≤ 30 min), missingness flag
CGM lags, slopes, acceleration, variability
Steps 15–120 min, HR vs 24-h baseline
HRV vs personal 3-day baseline (stress)
Last completed night: hours, efficiency, deep %
Logged carbs / glycaemic load 30–180 min

Leakage-safe by construction
(only data available at time t; unit-tested)

Evaluation protocol

Subject split 60 / 20 / 20 (train / conformal / test)
Twin calibrated on days 1–10 only
Metrics on days 11–14 of unseen patients
External check: CGMacros (real participants)

prior · *CGM* · *features*

3 · DIGITAL TWIN CORE

1 PERSONALISE · physiological twin

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G' = −(SG + X + A)·G + SG·Gb + Ra(t)/V
X' = p2·(kx·max(G − Gb, 0) − X)
A' = (k_ex·cadence − A)/15
```

Ra: per-meal gamma absorption, τ from food macros

MAP calibration of 6 parameters per patient
(Gb, SG, kx, p2, gut speed, exercise sensitivity)
Nelder–Mead · multi-horizon loss + EHR prior

2 SYNCHRONISE · state observer
state nudged toward every CGM reading

twin state & trajectories

3 PREDICT · hybrid ML layer

Inputs: CGM + wearables + meal log + EHR
+ twin-simulated trajectories, carbs-on-board, insulin action

LightGBM Δglucose @ 30 / 60 / 120 min
Conformalised quantile regression → 90% intervals

Event classifier: P(>180 mg/dL in 2 h)
Threshold set for ≤ 1 false alert / patient-day

TreeSHAP explanations, grouped by data stream
translated to clinician language

4 SIMULATE · what-if engine

Roll the synchronised twin forward 5 h with planned
meals, portions and walks → peak, time > 180
Ranks region-appropriate meal swaps per patient

4 · CLINICAL APPLICATION

FastAPI service

/ward · /patients/{id} · /whatif
/alerts · /timeline · /meal-ranking · /fhir
Clinical service layer: TIR, GMI, CV,
food response, sleep effect, summaries

5 ACT · clinician dashboard

- Overview: KPIs, risk queue, triage map
- Observed + ML forecast + twin curve
- "Why this alert": TreeSHAP drivers
- Replay scrubber over the whole record
- Twin state: observation → projection
- What-if simulator + best meal swaps
- Alert history with real outcomes
- Wearables, food response, sleep, labs
- FHIR record · pre-consultation summary

Shared replay clock · dark/light ·
charts bundled; system-font fallback

Users

Endocrinologist / GP: OPD & tele-consult
Diabetes educator: diet counselling
Roadmap: patient app alerts (Hindi / Kannada),
ASHA-worker low-bandwidth mode,
ABDM HIP/HIU consent integration

PRIVACY & DEPLOYMENT

Synthetic / open anonymised data only (DPDP Act 2023, HIPAA sandbox rules) · ABHA masked in UI · purpose-limited fields · single Docker container, runs on-premise · MIT-licensed code · decision support, not a medical device

Data flow: EHR seeds the twin's prior → wearable stream synchronises it every 5 minutes → ML layer corrects what physiology misses (sleep debt, stress, dawn effect) → alerts with reasons and what-if simulations reach the doctor.

Stack: Python 3 · NumPy / SciPy / pandas · LightGBM · scikit-learn · FastAPI + Pydantic · HTML/CSS/JS + Plotly · HL7 FHIR R4 · Docker · pytest

Static stream (EHR) · Dynamic stream (wearables) · Mechanistic twin · Machine learning · Clinical application